Ayush Krishna Garg

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Technical Profiles

GitHub: github.com/Ayush-K-Garg — **LeetCode:** diuk3VAt95 (Contest Rating: 1828)

Education

National Institute of Technology, Jamshedpur **Aug 2023 – May 2027**
B.Tech in **Electronics and Communication Engineering** **CGPA: 8.40/10**

The Chintels School, Kanpur **Mar 2021 – Mar 2023**
Class 12 (ISC): **96.5%** — Class 10 (ICSE): **97.0%**

Technical Skills

Languages: C++, JavaScript, Python, SQL
Frameworks & Libraries: Flutter, Node.js, Express.js, Socket.IO, FastAPI, React.js, LangChain
AI & ML: Generative AI, RAG, FAISS Vector Search
Systems & Infrastructure: Distributed Systems, Multithreading, TCP/UDP Sockets, System Design
Databases: MongoDB, MySQL, Firebase Firestore, PostgreSQL, Redis
Tools & DevOps: Docker, Git, GitHub, Postman, JMeter, Android Studio, VS Code

Projects

Distributed File Storage System | ***C++, Docker, TCP/UDP, Tailscale*** GitHub **Mar 2026**

- Engineered a robust **Master-Slave distributed storage cluster** acting as a private cloud with seamless node-to-node TCP communication and automatic node discovery via **UDP heartbeats**.
- Leveraged a custom **C++ thread pool** and parallel chunking to maximize hardware utilization, achieving a high-speed data throughput of **60 MB/s** during concurrent I/O operations.
- Implemented fault tolerance using **RF=2 replication** and end-to-end **SHA-256 checksum** verification to prevent data corruption and ensure high availability across multiple storage nodes.

CodeExplain – Repository Tutor | ***Python, FastAPI, React.js, LangChain, FAISS, Gemini API*** GitHub Live **May 2026**

- Built a **RAG codebase tutor** ingesting GitHub repos/ZIPs, parsing source code with **AST-aware splitters** (preserving class/function scopes), and storing **768-dim embeddings** in a persistent **FAISS** index via a 3-tier **FastAPI** backend.
- Optimized backend memory to stay below **100 MB RAM** (512 MB host limit) by processing embeddings in **15-chunk batches**; used **shallow Git cloning (depth=1)** to cut repository download times.
- Resolved regional Google SDK routing failures (`v1beta` 404s on cloud nodes) by using a custom `GeminiEmbeddings` REST client targeting the stable `/v1/` API, achieving **100% platform uptime**.

Student Portal App | ***Flutter, InAppWebView, Webscraping*** GitHub App **Sep 2025**

- Developed a cross-platform Flutter utility adopted by **400+ students** for automated NIT Jamshedpur SAP portal access, reducing login and navigation time by roughly **90% (from 120s to 10s)**.
- Engineered a **security bypass layer** using JavaScript injection and User-Agent spoofing to bypass legacy portal barriers and automate complex authentication sequences.
- Integrated **SharedPreferences** for encrypted local credential persistence and request caching to ensure reliable performance on unstable campus networks.

TrivIQ – Real-Time Multiplayer Quiz | ***Node.js, Express.js, Socket.IO, MongoDB, Firebase, Flutter*** GitHub Live **Jun 2025**

- Built a full-stack real-time multiplayer platform using **WebSockets (Socket.IO)** for bidirectional communication, maintaining low latency for concurrent global users.
- Architected a **room-based backend** with strict game-state isolation, enabling simultaneous quiz sessions with live leaderboard synchronization and Firebase-backed user authentication.

Achievements

- Semi-finalist in **HackOn with Amazon 6.0**, ranking among the **top 300 teams** selected from **70,265+ registrations**.
- **Problem Solving:** Solved **800+ problems** across LeetCode and Codeforces; achieved Global Rank **2,147** (LeetCode) among 27,350 participants in Weekly Contest 501, Global Rank **1,439** in Contest 504, and Global Rank **7,313** (Codeforces) in Div. 2 rounds.
- **JEE Main 2023:** Secured **AIR 13,555** (**98.84 percentile**) among 1.2 million candidates; qualified for **JEE Advanced 2023** with a rank of **20,582**.

Leadership Experience

Head of App Development – SECE, NIT Jamshedpur **Aug 2024 – Present**

- Led a 3-member team building and deploying **2 cross-platform applications** on the **Indus App Store** serving **200+ students**, while mentoring **juniors** in Flutter, Git workflows, code reviews, and full-stack development practices.